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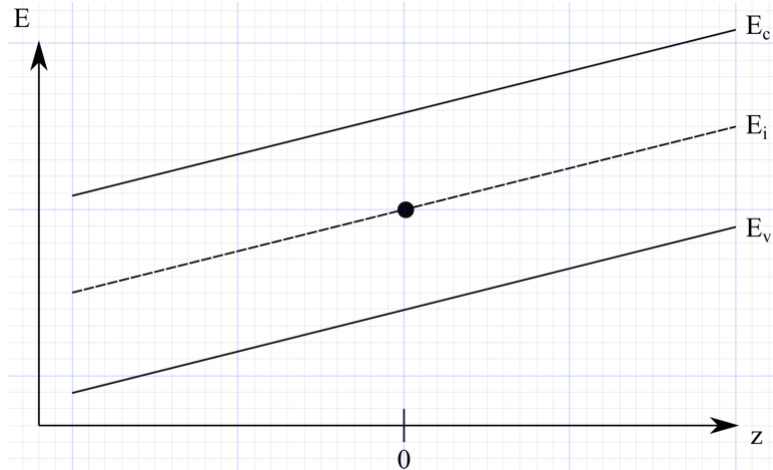
For all problems use

$n_i = 10^{10} / \text{cm}^3$ @ $T=300\text{K}$ for Si, $kT = 0.026 \text{ eV}$ at $T=300\text{K}$, $q=1.602\text{e-}19\text{C}$, $\epsilon_0 = 8.854\text{e-}14 \text{ F/cm}$, $\epsilon_{\text{Si}} = 11.8\epsilon_0$, $\epsilon_{\text{ox}} = 3.9\epsilon_0$, $E_{\text{G,Si}} = 1.12 \text{ eV}$, $E_{\text{G,Ge}} = 0.66 \text{ eV}$.

1. (10 pts) A Si sample is doped with 10^{17} phosphorous (P) atoms / cm^3 . What are the electron and hole concentrations, n and p , at $T=300\text{K}$?

2. Consider the following equilibrium band diagram for a portion of a semiconductor sample with a built-in electric field at $T = 300 \text{ K}$. The electron and hole effective masses are both equal to $0.3m_0$, where m_0 is the bare electron mass, and $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$.

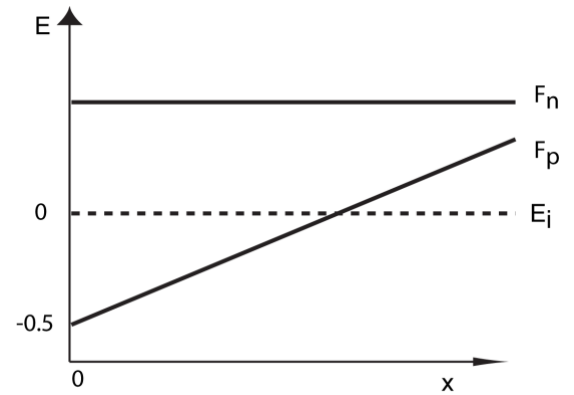
10 pts each



- Sketch the Fermi level as a function of position through the indicated point across the width of the band diagram above.
- On the band diagram, sketch the direction of the electric field.
- Show with an arrow labeled ($J_{\text{N-diffusion}}$) the direction of the electron diffusion current.
- Show with an arrow labeled ($J_{\text{H-diffusion}}$) the direction of the hole diffusion current.
- Show with an arrow labeled ($J_{\text{H-drift}}$) the direction of the hole drift current.
- Show with an arrow labeled ($J_{\text{N-drift}}$) the direction of the electron drift current.
- What is the direction of the total current? Why?
- What is the electron density (in cm^{-3}) at $z=0$?
- What is the hole density (in cm^{-3}) at $z=0$?

3. The intrinsic Fermi level, E_i , and the quasi-Fermi levels are shown at right.

(a) (10 pts) Is there an electron current? Yes / No.
If 'Yes', determine its direction and whether it is a result of drift or diffusion. Explain with an equation.



(b) (10 pts) Is there a hole current? Yes / No. If 'Yes', determine its direction and whether it is a result of drift or diffusion. Explain with an equation.

4. (20 pts) Your company makes a n^+p diode. They want to double its current. Which do you suggest? Why? Explain using equations.

- (a) doubling the donor doping
- (b) halving the donor doping
- (c) doubling the acceptor doping
- (d) halving the acceptor doping

5. (10 pts) An abrupt pn junction with a built-in voltage of 0.7 V has a junction capacitance of 2 pF at zero bias. What bias must be applied to obtain a 1 pF capacitance (include both magnitude and the sign of the bias)?

6. (10 pts) Your company is having trouble making an ohmic contact to a layer of n-type Si doped $10^{16} / \text{cm}^3$. What would you suggest?

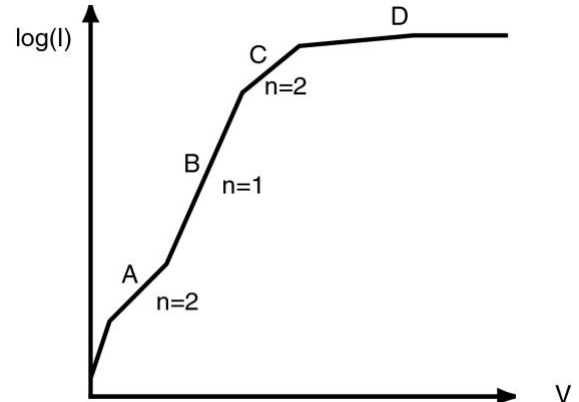
7. Below is a log plot of the current versus voltage of a pn diode. 'n' is the ideality factor. Describe the dominant current mechanism in regions (10 pts each)

(a) A

(b) B

(c) C

(d) Explain why the curve flattens out in region D

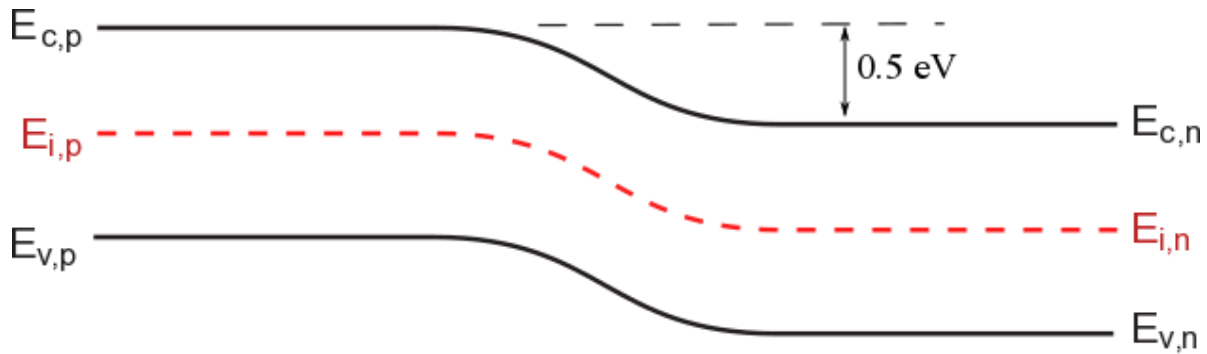


(e) Which would be the more likely semiconductor from which this diode was made, Si or Ge? Why?

8. (40 pts) Consider a Si diode biased in region B in the figure above. For automotive applications, integrated circuits must operate in the temperature range of -40°C to $+125^{\circ}\text{C}$. Assume that the diffusion coefficients, lifetimes, and bandgap are unaffected by the temperature change. If the applied voltage is fixed at $V_A = 0.7\text{ V}$, what is the ratio of the max current to the min current over this temperature range? (Convert to Kelvin before calculating numbers)

9. (10 pts) For a pn junction diode, describe two mechanisms of reverse bias breakdown.

10. Below is the band diagram of an ideal Si diode under forward bias of 0.3 V. It consists of an abrupt pn junction, $N_A = N_D$, and $T = 300$ K.



(a) (10pts) Determine N_A and N_D ($1/\text{cm}^3$).

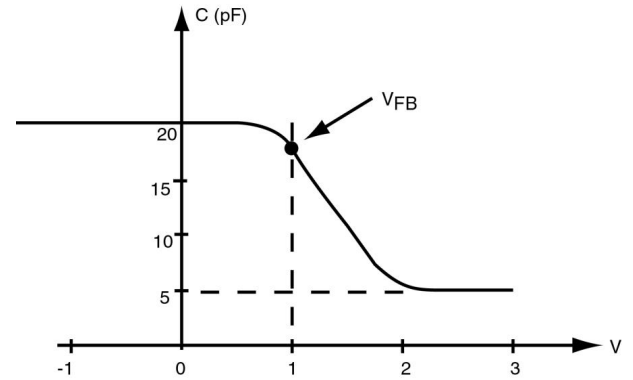
(b) (10pts) Determine $E_{i,p} - F_p$ on the p-side and $F_n - E_{i,n}$ on the n-side.

(c) (20pts) On the band diagram, draw and label the quasi-Fermi levels throughout the device under the assumptions leading to the ideal diode equation. Indicate the depletion region with dashed lines.

(d) (30 pts) Determine the maximum electric field.

11. At right is the CV characteristic for a Si MOS capacitor with a gate area of 10^{-3} cm^2 .

(a) (10 pts) Given $\phi_{ms} = 0.2 \text{ V}$, find Q_{ss}/q (cm^{-2}).



(b) (10 pts) What is the minimum depletion capacitance, C_{dt} , in (pF).

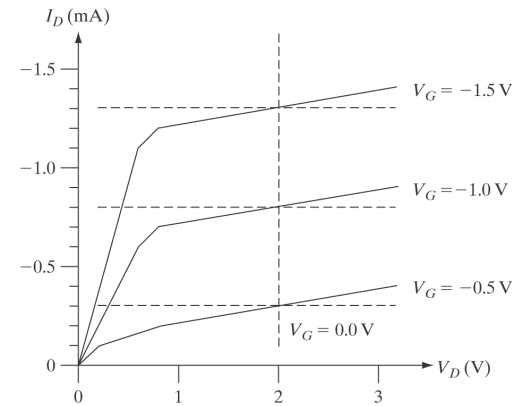
(c) (10 pts) Is the CV curve a high-frequency or low-frequency curve? Why?

(d) (10 pts) If this MOS capacitor is actually the gate of a MOSFET with the source and drain grounded and $V_G > V_t$, sketch the CV curve on the plot above. Briefly explain.

(e) (20pts) For the MOSFET of (d) at $T = 300 \text{ K}$, what is the inverse subthreshold slope at threshold (in units of mV/dec)?

(f) (10pts) For this same FET, assuming this value of S is fixed throughout the subthreshold region, how much gate voltage is required to reduce the drain current by a factor of 10^6 ?

12. (10 pts) Is the I-V characteristic at right from a long-channel or short channel device? Explain.



13. (10 pts each) For an n-channel MOSFET (p-type substrate) with a Si / SiO₂ interface charge of $Q_{ss} = -1.602 \times 10^{-9} \text{ C/cm}^2$, what is the effect of decreasing the oxide thickness on the following; circle one for each, and support answer with an equation:

(a) The flat-band voltage. increase decrease no change

(b) The threshold voltage. increase decrease no change

(c) The inverse subthreshold slope increase decrease no change

14. (10 pts) What are the primary effects of DIBL. Explain.

15. (10 pts) What is the primary effect of GIDL. Explain.

16. (20 pts) Modern integrated circuits on-chip temperatures reach 100°C. At this temperature, if you want to turn-off the current by a factor of 10^5 in a FET. What is the absolute theoretical minimum change in gate voltage that is required?

17. (10 pts) For ultra-scaled FETs with gate lengths of 5 nm or less, what is the primary cause of off-state leakage current? Explain and illustrate with a band diagram.

18. (40pts) A FET built with a new material in a research lab will have the following typical parameters: $t_{ox} = 300$ nm of SiO₂, $V_t = 20$ V, $L = 4$ μm , and $W = 2$ μm .

With $V_{GS} = 24$ V and $V_{DS} = 2$ V, you measure $\frac{\partial I_D}{\partial V_{GS}} = 1.15$ $\mu\text{A/V}$. What is the mobility of the material?

19. (10pts each) Label each figure in the space provided as corresponding to one of the following states: accumulation, weak-inversion, depletion, strong inversion, flat-band, or threshold.

(10 pts each) Label each state (a)-(f) on the CV curve at the bottom.

